

Melvin Chirag Karupati

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EDUCATION

Eastern Michigan University

Bachelor of Science in Computer Science

Ypsilanti, MI

Expected April 2027

- **Concentration:** Pursuing a specialized concentration in Artificial Intelligence and Machine Learning, focusing heavily on modern algorithms and deep learning architectures.
- **Achievements:** Demonstrated consistent technical excellence by securing two separate hackathon victories in competitive regional coding marathons against hundreds of participants.

PROFESSIONAL EXPERIENCE

Student Assistant

Aug 2024 – Present

Disability Resource Center (DRC) / CATE Lab

Ypsilanti, MI

- Manage technical and web accessibility updates for disability resources, ensuring strict compliance standards are met while effectively communicating architectural changes to various university stakeholders.
- Diagnose and resolve complex user-facing software issues on a daily basis, maintaining a highly responsive technical support environment for both students and faculty across the campus network.

TECHNICAL PROJECTS

Osiris | Python, PyTorch, V-JEPA, OpenCV, MediaPipe, FastAPI

- Architected an intelligent computer vision system focused on comprehensive physical-world understanding, utilizing real-time American Sign Language to speech translation as the initial applied use case.
- Built a custom gesture recognition perception layer by leveraging the MediaPipe library to accurately map and assign programmed actions to complex real-world physical gestures.
- Engineered a robust representation layer utilizing PyTorch and V-JEPA to extract 1,000-dimensional embedding vectors, establishing a foundation to eventually integrate advanced VL-JEPA capabilities.
- Developed Layer 3 decision logic using cosine similarity against a database, supported by a scalable FastAPI and pgvector backend for ultra-low latency visual processing and complex scene retrieval.

EventsOS | React, TypeScript, FastAPI, Gemini, MongoDB

- Co-developed a sophisticated multi-agent event operations platform that successfully secured the MLH Best Use of Gemini API award during the highly competitive GrizzHacks 8 hackathon.
- Engineered the dynamic client-facing user interface and seamlessly integrated the Gemini API to intelligently route unstructured user prompts across a distributed network of specialized domain agents.
- Orchestrated real-time operational workflows to manage concurrent task execution across event planning, sponsorship tracking, and compliance, significantly enhancing overall platform efficiency.

Lingo | React, Next.js, Tailwind CSS, FastAPI, Gemini

- Built an innovative AI cultural translator capable of identifying complex idioms and regional slang, dynamically rewriting them into culturally neutral language to secure a victory at SpartaHack 11.
- Implemented a highly responsive modern web UI and intricate LLM workflow designed to instantly surface phrase definitions, contextual nuances, and actionable suggestions for both senders and receivers.
- Constructed a robust backend processing pipeline utilizing FastAPI to efficiently handle concurrent user requests and seamlessly parse structured language outputs from the Gemini model.

Otto_sys | Python, Claude MCP, Agent Architecture

- Developed a sophisticated human-gated, domain-bounded automation framework designed to autonomously evaluate and delegate complex operational tasks to a network of specialized programmatic sub-agents.
- Securely integrated Model Context Protocol (MCP) servers into the architecture to ensure absolute separation between core orchestration logic and external tool execution capabilities.

Manas | JavaScript, HTML/CSS, WebGL, UI State Management

- Designed and constructed an interactive 3D astrophysics canvas featuring terminal-driven scenarios, live data telemetry, and highly modular visual simulations by leveraging the rendering power of WebGL.
- Engineered advanced UI state management protocols to ensure flawless synchronization between the complex 3D viewport rendering and the underlying JavaScript event-driven architecture.

LEADERSHIP & INVOLVEMENT

Hack the Nest (Eastern Michigan University)

November 2026

Finance Lead & Co-Organizer

Ypsilanti, MI

- Spearheading the comprehensive financial planning, budget allocation, and strategic resource management for a large-scale, 200+ attendee computer science hackathon.
- Collaborating actively with cross-functional organizational teams to secure substantial corporate sponsorships and effectively manage logistical resources for technical infrastructure.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, HTML/CSS, MATLAB.

AI & Computer Vision: PyTorch, OpenCV, MediaPipe, V-JEPA, VL-JEPA, Cosine Similarity, Gemini.

Frontend & UI Design: React, Next.js, Tailwind CSS, Glassmorphism, Three.js (WebGL), GSAP.

Backend & Data: FastAPI, Supabase, pgvector, MongoDB, REST APIs, Node.js, Claude MCP.

Developer Tools: Git/GitHub, Docker, Vite, WebRTC, Arduino, JUnit, System Testing, Object-Oriented Design.